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Accurate Estimation of Oil Well Flow Rates Through Integrated Numerical Modeling of Pressure and Temperature Profiles in the Wellbore

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Abstract

Accurate flow rate estimation is critical for effective oil well management, production optimization, and reservoir performance evaluation. Traditional measurement techniques—such as Venturi meters, multiphase separators, and other direct sensing devices—though widely used, are often costly, maintenance-intensive, and operationally complex. This paper introduces an advanced numerical modeling framework that leverages both pressure and temperature data to improve flow rate prediction accuracy, consistency, and adaptability across various well conditions. Unlike previous approaches that rely exclusively on either pressure or temperature, the proposed method explicitly couples the thermal and hydraulic behavior of fluids in the wellbore. The model simultaneously solves the continuity, momentum, and energy conservation equations using finite difference methods. It accounts for critical physical effects such as heat transfer, geothermal gradients, Joule–Thomson cooling, and multiphase flow characteristics, allowing a more realistic simulation of the dynamic wellbore environment.

Field data from three producing oil wells are used to evaluate and compare multiple modeling strategies: pressure-only, temperature-only, combined pressure–temperature, and dimensionless approaches incorporating different thermal assumptions. The results demonstrate that integrating pressure and temperature data within a unified numerical framework significantly improves flow rate prediction accuracy. Moreover, applying dimensionless normalization enables better comparison across variables with different magnitudes and units, enhancing model stability and generalizability. Sensitivity analysis indicates that the integrated model adapts well to varying operational scenarios and thermal profiles, outperforming conventional temperature-based graphical methods. Although pressure data generally yield more robust results due to higher sensor reliability and a stronger correlation with flow dynamics, incorporating temperature data refines predictions and enhances interpretability, especially in complex or deep wells.

This study contributes a practical and scalable methodology for non-intrusive flow rate estimation, reducing dependence on complex physical metering systems. The findings suggest that integrated pressure-temperature modeling—when supported by dimensionless scaling and iterative calibration—is a

promising advancement for real-time digital oilfield monitoring, offering enhanced accuracy, reliability, and operational efficiency.

Introduction

Flow rate measurements are crucial for effective reservoir management, production optimization, and ensuring operational efficiency. Although direct flow rate measurement methods, such as multiphase separators, Venturi meters, and orifice plates, provide accurate results, they typically involve substantial logistical complexity, significant maintenance requirements, and high operational costs, particularly in remote or harsh environments. As a result, indirect numerical modeling approaches have emerged as practical, cost-effective alternatives, facilitating continuous and reliable flow rate estimation.

Historically, many indirect methods have primarily focused on pressure-based approaches, due to their direct relationship with fluid dynamics and ease of implementation across various reservoir conditions. Similarly, temperature-based methods have been widely analyzed, leveraging heat exchange between wellbore fluids and the surrounding geological formations. Extensive research has been devoted to wellbore thermal modeling and its correlation with fluid flow rates.

Early studies (Nowak, 1953; Schorre, 1954) laid the groundwork for temperature analysis in pipelines, with Ramey (1962) introducing a foundational model for wellbore temperature during injection and production. Subsequent enhancements addressed phase changes (Satter, 1965; Willhite, 1967), Joule–Thomson effects (Steffensen & Smith, 1973), and multiphase flow (Sagar et al., 1991). Simplifications and analytical models followed (Shiu & Beggs, 1980; Hasan & Kabir, 1994). Recent efforts (Izgec et al., 2010; Nian et al., 2015) have focused on integrating pressure and temperature data for more accurate flow rate estimation, while highlighting challenges such as sensor limitations and thermal property variability. Sadri et al. (2019) developed a compositional thermal multiphase wellbore model that accounts for non-isothermal effects to enhance flow prediction accuracy in gas-lifted wells.

Recent studies have adopted advanced approaches such as machine learning and real-time data analysis for flow rate prediction. Garcia et al. (2020) and Ibrahim et al. (2022) demonstrated the potential of machine learning models, while noting challenges in data adaptability and retraining. Grelli et al. (2023) and Izgec (2023) advanced real-time virtual flow metering, highlighting benefits in cost efficiency alongside the need for robust data and computational resources.

Igder et al. (2024) introduced a graphical method based on temperature data and enthalpy balance to estimate oil well production rates. While effective and user-friendly, its reliance on thermal data alone limited its accuracy under varying pressure conditions. To address this, the present study proposes an advanced numerical modeling approach that integrates both pressure and temperature data within a unified framework. This method captures the coupled hydraulic-thermal interactions often overlooked in earlier models, significantly enhancing prediction accuracy. By applying dimensionless data normalization, it also mitigates scale disparities between pressure and temperature measurements. The study aims to assess the predictive performance of this integrated model across diverse well conditions, and to compare its strengths and limitations against traditional single-parameter approaches.

Numerical Modeling Methodology

Unlike the temperature-only graphical method by Igder et al. (2024), this numerical approach explicitly couples pressure and temperature data through governing equations, capturing their interdependencies and enhancing model accuracy across diverse operational scenarios. The model solves continuity, momentum, and energy conservation equations via finite difference methods. To calculate pressure, a numerical model integrates pressure gradients along the wellbore by dividing it into computational intervals where the pressure gradient is assumed constant. Flow patterns and liquid holdup—key to pressure drop in two-phase flow—are determined during this process using the mechanical modeling approach of Ansari et al. (1994),

which links hydrodynamic models to specific flow regimes. Temperature predictions incorporate fluid thermodynamics, geothermal gradients, and heat transfer. An iterative algorithm refines coupled pressure and temperature profiles from bottom-hole to surface until convergence, improving flow rate estimates.

Governing Equations

This study presents a comprehensive numerical model that simultaneously solves the continuity, momentum, and energy equations to couple pressure and temperature profiles within the wellbore. Unlike previous temperature-only approaches, the model explicitly accounts for pressure variations and their impact on temperature gradients via the Joule-Thomson effect. The coupled differential equations are discretized using finite difference methods, allowing iterative numerical solutions with improved accuracy. As reservoir fluids flow upward, pressure losses occur due to friction, gravity, and acceleration, influenced by flow rate, temperature, fluid properties, and wellbore geometry. Pressure loss is derived from momentum conservation and modeled using the mechanistic approach of Ansari et al. (1994) for flow pattern identification, liquid holdup, and pressure drop estimation. While detailed pressure loss analysis is beyond this study's scope, pressure calculations support temperature predictions. Heat transfer between the fluid and surrounding formation further affects temperature, governed by mass flow rate and thermophysical properties. The energy balance for the control volume Fig. 1 is expressed as follows:

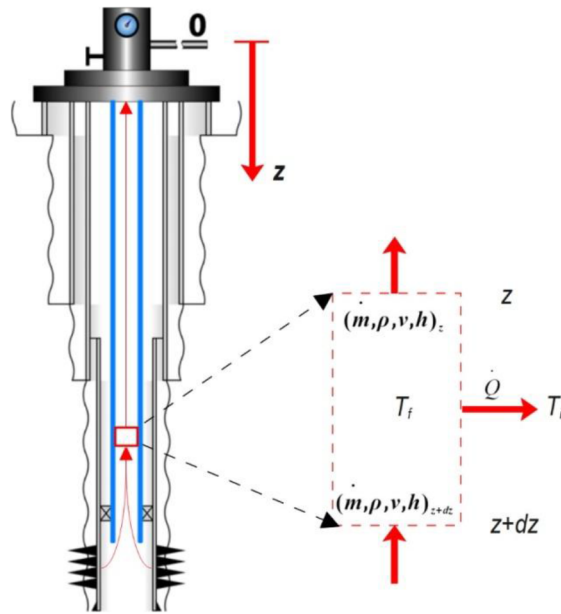


Figure 1—Flow path and enthalpy balance within the control volume for a typical flowing wellbore assuming no-entry zone.

$$\dot{m} \left(\hat{h} + \frac{1}{2} v^2 + gz \right)_{in} - \dot{m} \left(\hat{h} + \frac{1}{2} v^2 + gz \right)_{out} - \dot{Q} - \dot{W} = 0 \quad (1)$$

Where $\dot{m}$ is mass flow rate, $\hat{h}$ is the specific fluid enthalpy and $\dot{Q}$ is the rate of heat loss from well column fluid to the formation. The $\dot{Q}$ in Eq. 1 equation can be calculated as follows:

$$\dot{Q} = P U_{to} (T_f - T_{ei}) dz \quad (2)$$

Where P is tubing outside perimeter is, U_{to} is overall heat transfer coefficient based on tubing outside diameter, T_f is fluid temperature and T_{ei} is formation temperature. Substituting Eq. (2) in Eq. (1) and dividing it by dz results in the following equation in differential form:

$$\dot{m} \frac{d \left(\hat{h} + \frac{1}{2} v^2 + gz \right)}{dz} - P U_{to} (T_f - T_{ei}) = 0 \quad (3)$$

When there is negligible phase change, enthalpy is a function of pressure and temperature and is given by:

$$d\hat{h} = \left(\frac{\partial \hat{h}}{\partial T} \right)_p dT + \left(\frac{\partial \hat{h}}{\partial p} \right)_T dp = c_p dT - c_p c_J dp \quad (4)$$

Where c_J is Joule–Thomson coefficient, and c_p is the mean heat capacity of the fluid at constant pressure. Substitute Eq. 4 into Eq. 3 and expand the derivative:

$$\frac{dT}{dz} = \frac{PU_{to}(T_f - T_{ei})}{\dot{m}c_p} + c_J \frac{dP}{dz} - \frac{v}{c_p} \frac{dv}{dz} - \frac{g}{c_p} \quad (5)$$

The first term on the right-hand side represents heat transfer, accounting for the exchange of heat between the fluid and its surroundings. The second term is the pressure gradient term and reflects the effect of pressure variations on temperature, particularly significant in deep wells and compressible flows where the Joule–Thomson effect is prominent. The third term is the kinetic energy term and captures the impact of changes in fluid velocity on the temperature gradient, which is essential in high-velocity flows or regions with varying cross-sectional areas. Finally, the potential energy term accounts for the influence of elevation changes due to gravitational forces. For complex equations with coupled terms, numerical methods provide approximate solutions through discretization or iterative solving.

Auxiliary Heat Transfer Equations

Detailed heat transfer modeling requires calculating the overall heat transfer coefficient by the summation of a series of resistances. To solve Eq. (5), the overall heat transfer equation is incorporated as an auxiliary equation to complement the temperature gradient equation. Refer to Fig. 2. U_{to} can be determined as follows:

$$\frac{1}{U_{to}} = \left[\frac{r_{to}}{r_{ti} h_f} + \frac{r_{to} \ln(r_{to}/r_{ti})}{k_t} + \frac{1}{h_a} + \frac{r_{to} \ln(r_{co}/r_{ci})}{k_c} + \frac{r_{to} \ln(r_{wb}/r_{co})}{k_{cem}} \right] + [R_{cond,e}] \quad (6)$$

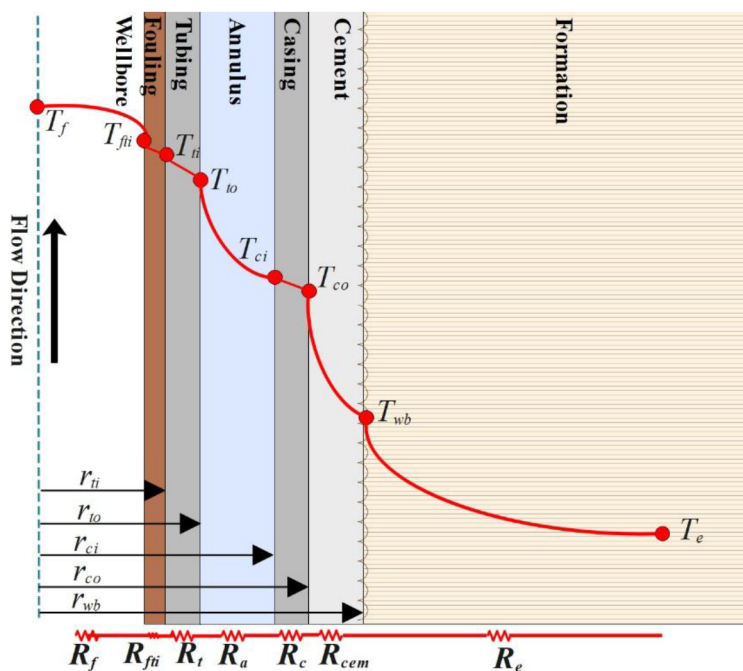


Figure 2—The development of the temperature distribution and heat resistances for the radial system around the cross section of typical flowing wellbore.

The right-hand side of Eq. (6) is the summation of a series of resistances, which are tubing fluid resistance, tubing resistance, annulus fluid resistance, casing resistance, cement resistance and earth resistance,

respectively. In contrast, the terms inside the second bracket account for transient heat transfer in the formation.

Heat transfer in tubing fluid is forced convection. Its heat transfer coefficient can be calculated using the [Dittus and Boelter \(1930\)](#) explicit correlation for turbulent flow as follows:

$$Nu = 0.023Re^{0.8}Pr^{0.3} \quad (7)$$

Three dimensionless numbers in [Eq. \(7\)](#) are Nusselt, Reynolds and Prandtl numbers. Note that for the Reynolds number it is necessary to estimate the rate as an initial guess. By guessing the initial rate and calculating the dimensionless numbers mentioned above, the fluid heat transfer coefficient can be determined. The other challenge is the calculation of the natural convection heat transfer coefficient in the annulus. The following correlation is recommended by [Dropkin and Somerscales \(1965\)](#) for natural convection between two short vertical plates:

$$h_a = \frac{0.049(Gr.Pr)^{1/3}Pr^{0.074}k_{an}}{r_{to}\ln(r_{co}/r_{ci})} \quad (8)$$

Where the Gasthof number, Gr , reflects the ratio of buoyancy forces to viscous forces:

$$Gr = \frac{(r_{ci} - r_{to})^3 g \rho_{an}^2 \beta (T_{to} - T_{ci})}{\mu_{an}^2} \quad (9)$$

Where, β , ρ_{an} and μ_{an} are the coefficients of thermal expansion, density and viscosity of annulus fluid, respectively. As mentioned, [Eq. \(8\)](#) has been developed basically for natural convection between two short vertical plates and cannot be used directly for actual annular concentric cylindrical long pipes. Therefore, [Hasan and Kabir \(1994\)](#) pointed out using $h_a/4$ as a correction for annulus heat transfer coefficient based on their experience. In this study, applying this correction based on actual field data is essential to avoid significantly overestimating the annulus heat transfer coefficient.

According to [Ramey \(1962\)](#), the thermal resistance of the earth, denoted as $R_{cond,e}$ can be determined using the transient-time function as follows:

$$R_{cond,e} = \frac{f(t)}{k_e} \quad (10)$$

The function $f(t)$ delineates the time-dependent heat dissipation in the surrounding earth is expressed as:

$$f(t) = \frac{1}{2} E_i \left(\frac{-r_{wb}^2}{4\alpha t} \right) \quad (11)$$

where t is the production time, r_{wb} is the wellbore radius and α is the earth diffusivity.

Procedure and Computational Approach

The wellbore is divided into computational intervals that capture key features like phase transitions and geometry changes. Pressure and temperature profiles are calculated iteratively from the bottom of the well upwards, starting with known values such as bottom hole pressure and temperature, and assuming a production rate. Initial estimates for pressure and temperature at the second node are derived from expected pressure drop and geothermal gradient. These values are refined using flow regime analysis and heat transfer coefficients, which are used to compute the temperature gradient and update temperature values. The pressure and temperature are solved in a coupled iterative loop until convergence is achieved. The model calculates root mean square error (RMSE) across pressure, temperature, and total error by comparing simulation results with actual data. Weighted RMSE values across different flow rates help identify the optimal flow rate by minimizing total RMSE. [Figure 3](#) illustrates the flow path and computational intervals used in the simulation. The total RMSE is calculated using defined equations:

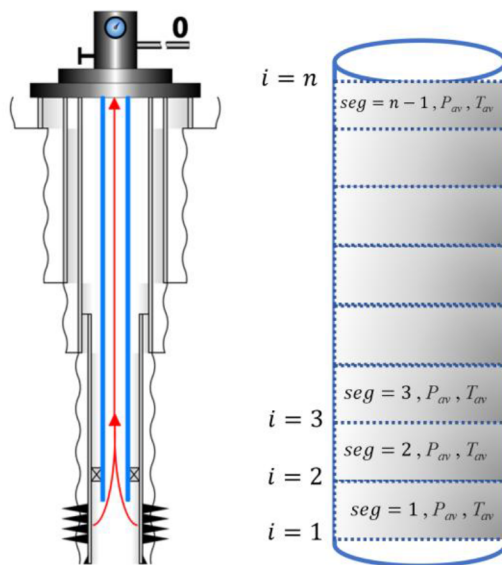


Figure 3—Flow path and schematic of computational intervals in a typical well.

$$RMS_{Total} = w_P * RMS_{Pressure} + w_T * RMS_{Temperature} \quad (12)$$

$$w_P + w_T = 1 \quad (13)$$

Where w_P and w_T are the weighting factors for pressure and temperature, respectively. When dealing with variables in different units and scales (e.g., pressure and temperature), it is essential to normalize and render the data dimensionless. This ensures accurate comparison and analysis by aligning the scales of the variables. The following equations are used for normalization:

$$P_D = \frac{P - P_{wh}}{P_{bh} - P_{wh}} \quad (14)$$

$$T_D = \frac{T - T_{wh}}{T_{bh} - T_{wh}} \quad (15)$$

To compare results and select the optimal flow rate, the dimensionless RMSE is used as follow:

$$RMS_{Dimensionless Total} = w_P * RMS_{Dimensionless Pressure} + w_T * RMS_{Dimensionless Temperature} \quad (16)$$

Results and Discussion

Numerical modeling results are compared with actual pressure, temperature, and flow rate data to evaluate accuracy. The key challenge is identifying whether pressure, temperature, or their combination provides the most accurate flow rate estimation under varying conditions. To address this, six numerical modeling approaches are considered: (1) modeling based on pressure, (2) modeling based on temperature, (3) combining pressure and temperature under a non-isothermal flow assumption, (4) using dimensionless pressure and temperature with a non-isothermal flow assumption, (5) using dimensionless pressure and temperature with a constant temperature assumption in the well column, and (6) using dimensionless pressure and temperature with a geothermal gradient assumption.

Field data from oil wells AZ30-Case1, AZ30-Case2, and AZ53-Case1 were analyzed using the six numerical modeling approaches, with operational, schematic, and thermo-physical parameters summarized in Table 1. The flow rate is iteratively refined to minimize the root mean square error (RMSE), until the optimal flow rate is identified.

Table 1—Well, fluid and thermophysical field data.

Well	AZ30 Case 1	AZ30-Case2	AZ53-Case1
Oil flow rate, STB/D	421	2494	1194
GOR, SCF/	440	440	212
Water-cut	≈ 0	≈ 0	≈ 0
True vertical depth, ft	10017	10017	12465
Tubing ID, in	2.992	2.992	2.992
Tubing OD, in	3.5	3.5	3.5
Casing ID, in	8.681	8.681	8.681
Casing OD, in	9.625	9.625	9.625
Tubing thermal conductivity, Btu/hr.°F.ft	26	26	26
Casing thermal conductivity, Btu/hr. °F.ft	26	26	26
Cement thermal conductivity, Btu/hr. °F.ft	0.5	0.5	0.4
Earth thermal conductivity, Btu/hr.°F.ft	1.4	1.4	1.16
Earth heat capacity, Btu/lbm. °F	0.22	0.22	0.22
Surface flowing temperature, °F	92.2	121	129.7
Bottom-hole temperature, °F	203	203	199
Geothermal gradient, °F /ft	0.0138	0.0138	0.0141
Oil API gravity	19	19	20
Gas specific gravity	0.88	0.88	0.9

Figure 4 illustrates the RMSE of pressure for various flow rates across all case studies and compares measured and simulated pressure data at optimal flow rates, showing excellent agreement. The calculated optimal flow rates and corresponding RMSE values are summarized in Tables 2, 3, and 4. Figure 5 illustrates the RMSE of temperature for various flow rates across all case studies and compares measured and simulated temperature data at optimal flow rates, showing excellent agreement. The calculated optimal flow rates and corresponding RMSE values are summarized in Tables 2, 3, and 4. Figure 6 illustrates the total RMSE for various flow rates across all case studies, based on combined temperature and pressure data. The optimal flow rates and corresponding RMSE values are summarized in Tables 2, 3, and 4.

Table 2—Optimal flow rate and error analysis using various approaches for Well AZ30-Case1 (actual flow rate = 421 STBD).

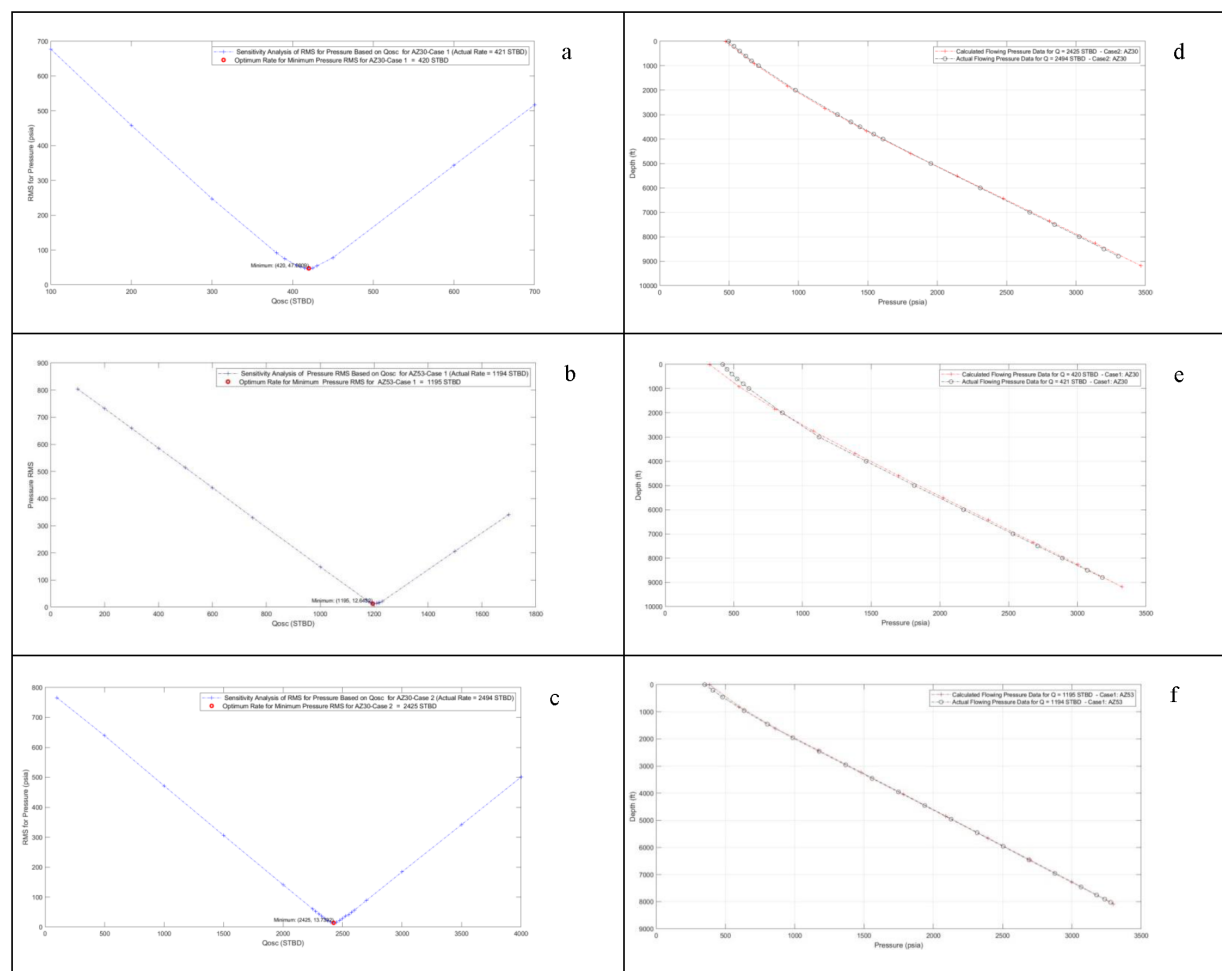
Model Type	Matching Basis	Modeling Assumptions	RMS					Predicted Flow Rate (STBD)	Relative Error (%)
			RMS_P psia	RMS_T °F	w_P	w_T	RMS Total (psia or °F)		
Numerical	Pressure Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	47.1	1.2	1	0	47.1	420	-0.2
Numerical	Temperature Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	51.6	1.1	0	1	1.1	410	-2.6
Numerical	Combined Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	47.1	1.2	0.5	0.5	24.1	420	-0.2
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	-	-	0.5	0.5	0.014	420	-0.2
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: constant temperature	-	-	0.5	0.5	0.18	417	-1.0
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: geothermal gradient	-	-	0.5	0.5	0.06	415	-1.4

Table 3—Optimal flow rate and error analysis using various approaches for Well AZ30-Case2 (actual flow rate = 2494 STBD).

Model Type	Matching Basis	Modeling Assumptions	RMS					Predicted Flow Rate (STBD)	Relative Error (%)
			RMS_P psia	RMS_T °F	w_P	w_T	RMS Total (psia or °F)		
Numerical	Pressure Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	13.7	3.3	1	0	13.7	2425	-2.8
Numerical	Temperature Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	35.9	3.2	0	1	3.2	2325	-6.8
Numerical	Combined Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	13.7	3.3	0.5	0.5	8.5	2425	-2.8
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	-	-	0.5	0.5	0.221	2425	-2.8
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: constant temperature	-	-	0.5	0.5	0.17	2360	-5.4
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: geothermal gradient	-	-	0.5	0.5	0.23	2330	-6.6

Table 4—Optimal flow rate and error analysis using various approaches for Well AZ53-Case1 (actual flow rate = 1194 STBD).

Model Type	Matching Basis	Modeling Assumptions	RMS					Predicted Flow Rate (STBD)	Relative Error (%)
			RMS_P psia	RMS_T °F	w_P	w_T	RMS Total (psia or °F)		
Numerical	Pressure Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	12.6	1.1	1	0	12.6	1195	0.1
Numerical	Temperature Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	15.1	1.1	0	1	1.1	1215	1.8
Numerical	Combined Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	12.6	1.1	0.5	0.5	6.9	1195	0.1
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: thermal model	-	-	0.5	0.5	0.01	1195	0.1
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: constant temperature	-	-	0.5	0.5	0.2	1225	2.6
Numerical	Combined Dimensionless Data	Pressure: pressure-based model considering reservoir and well column pressure drop Temperature: geothermal gradient	-	-	0.5	0.5	0.2	1235	3.4

**Figure 4—(a,b,c): RMSE based on pressure at different flow rates in Well AZ30-Case1, AZ30-Case2, AZ53-Case1 (d,e,f): measured pressure data and simulated pressure for all case studies at the optimal flow rate based on pressure.**

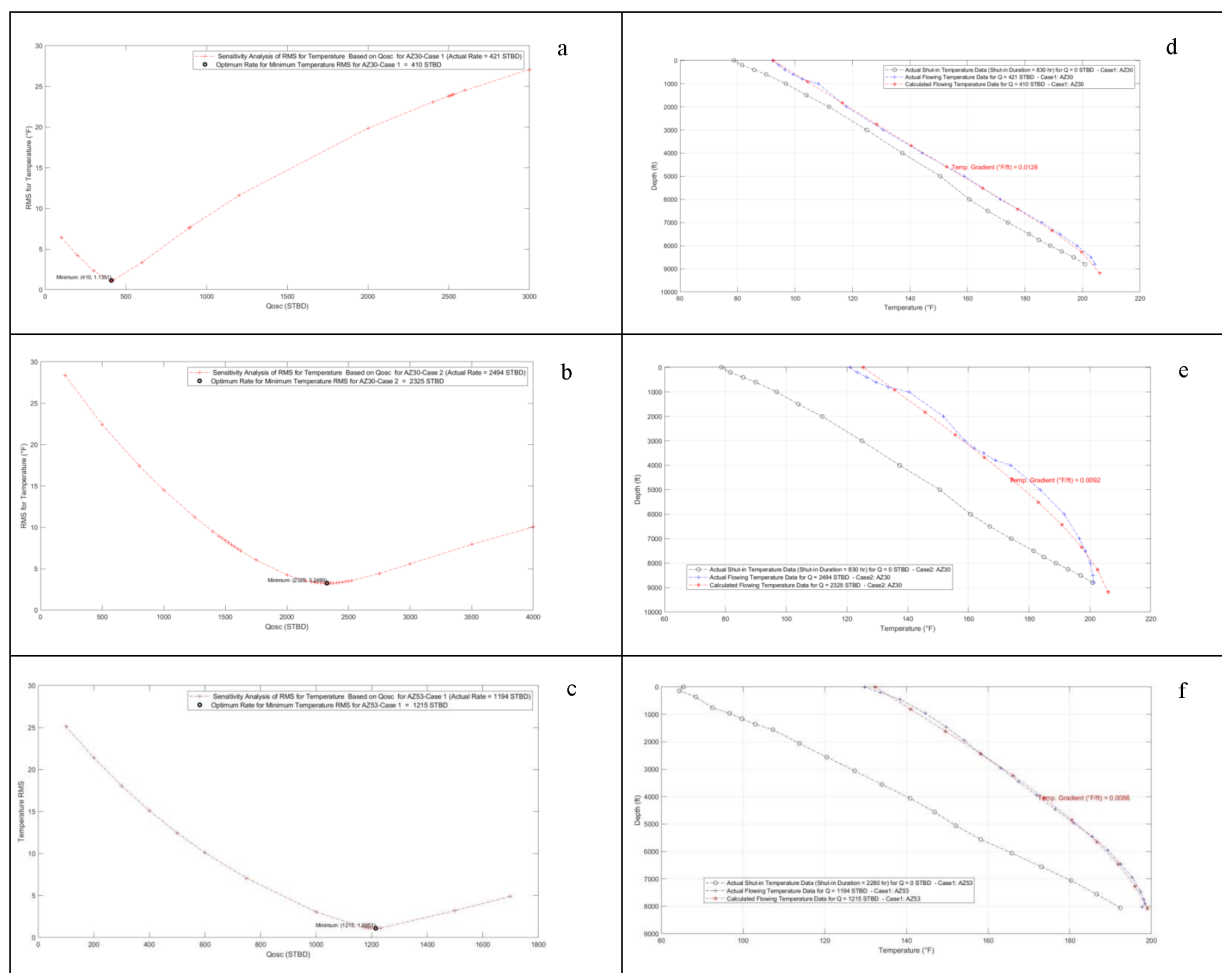


Figure 5—(a,b,c) RMSE based on temperature at different flow rates in Well AZ30-Case1, AZ30-Case2, AZ53-Case1 (d,e,f) measured temperature data and simulated temperature for all case studies at the optimal flow rate based on temperature.

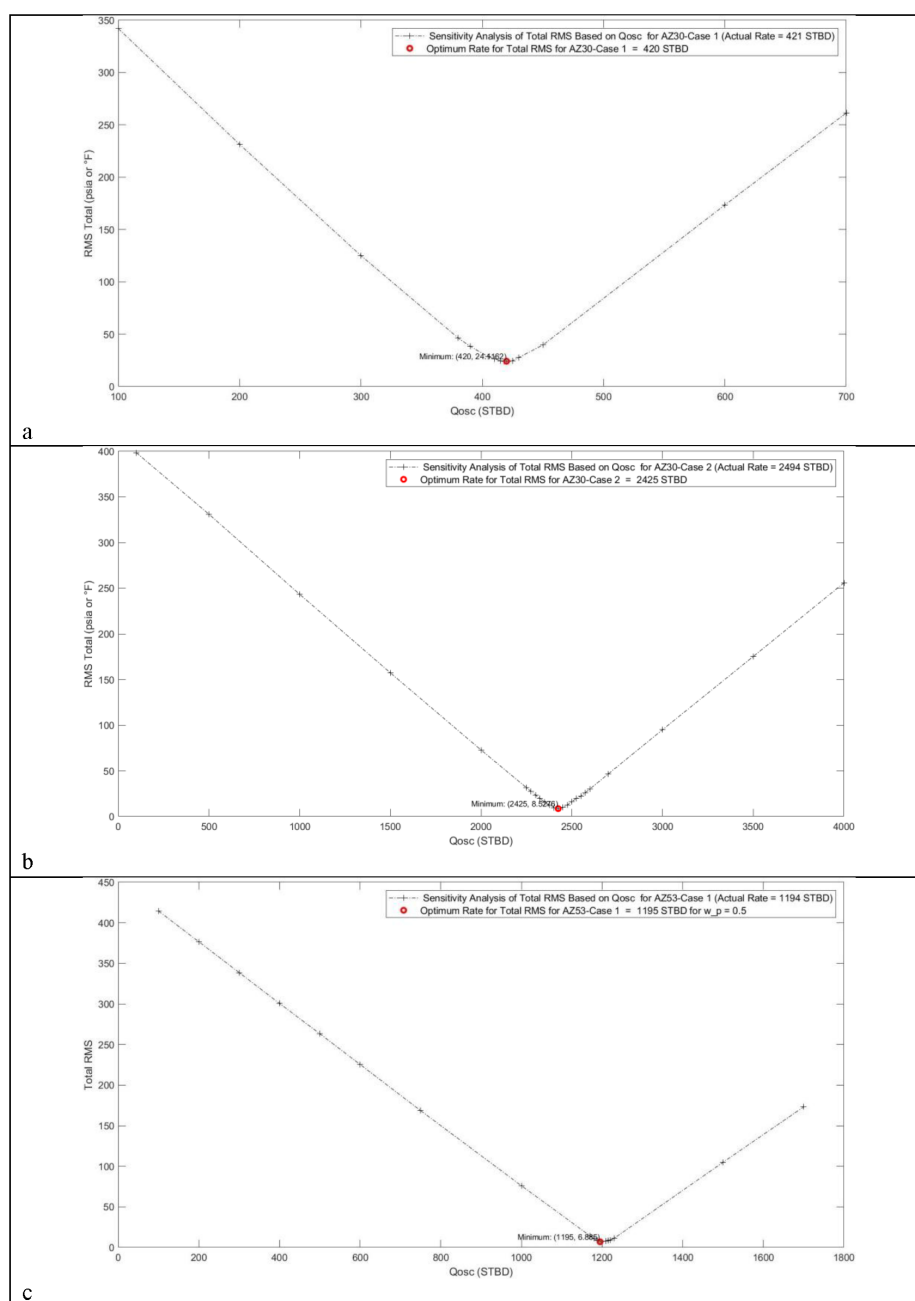


Figure 6—(a,b,c) Total RMSE for various flow rates in for Well AZ30-Case1, AZ30-Case2, AZ53-Case1 based on the combination of temperature and pressure.

When data were not dimensionless, applying weighting factors had little impact on the results due to the dominance of measured pressure values over temperature values. Even with high weighting for temperature, the combined model closely aligned with pressure-predicted results, demonstrating the model's inherent prioritization of pressure. Figure 7 illustrates the total dimensionless RMSE for various flow rates across all case studies, based on combined dimensionless temperature and pressure data. Figure 8 presents the total dimensionless RMSE under the assumption of constant well column temperature, using the average of bottom hole and wellhead temperatures. Figure 9 depicts the total dimensionless RMSE assuming temperature variation with the geothermal gradient. The optimal flow rates and corresponding dimensionless RMSE values for all cases are summarized in Tables 2, 3, and 4.

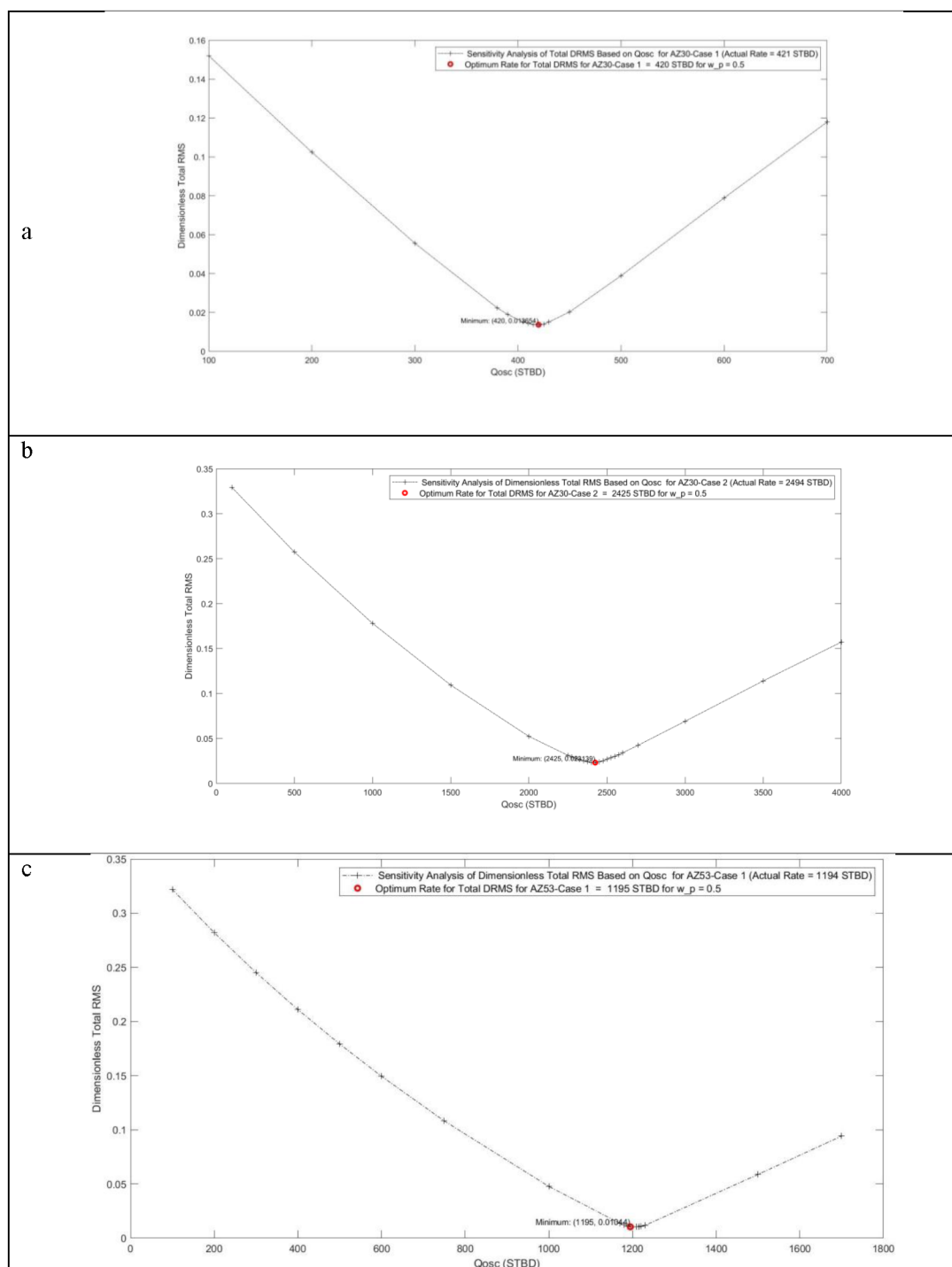


Figure 7—(a,b,c) Total dimensionless RMSE for various flow rates in for Well AZ30-Case1, AZ30-Case2, AZ53-Case1 based on the combination of dimensionless temperature and pressure.

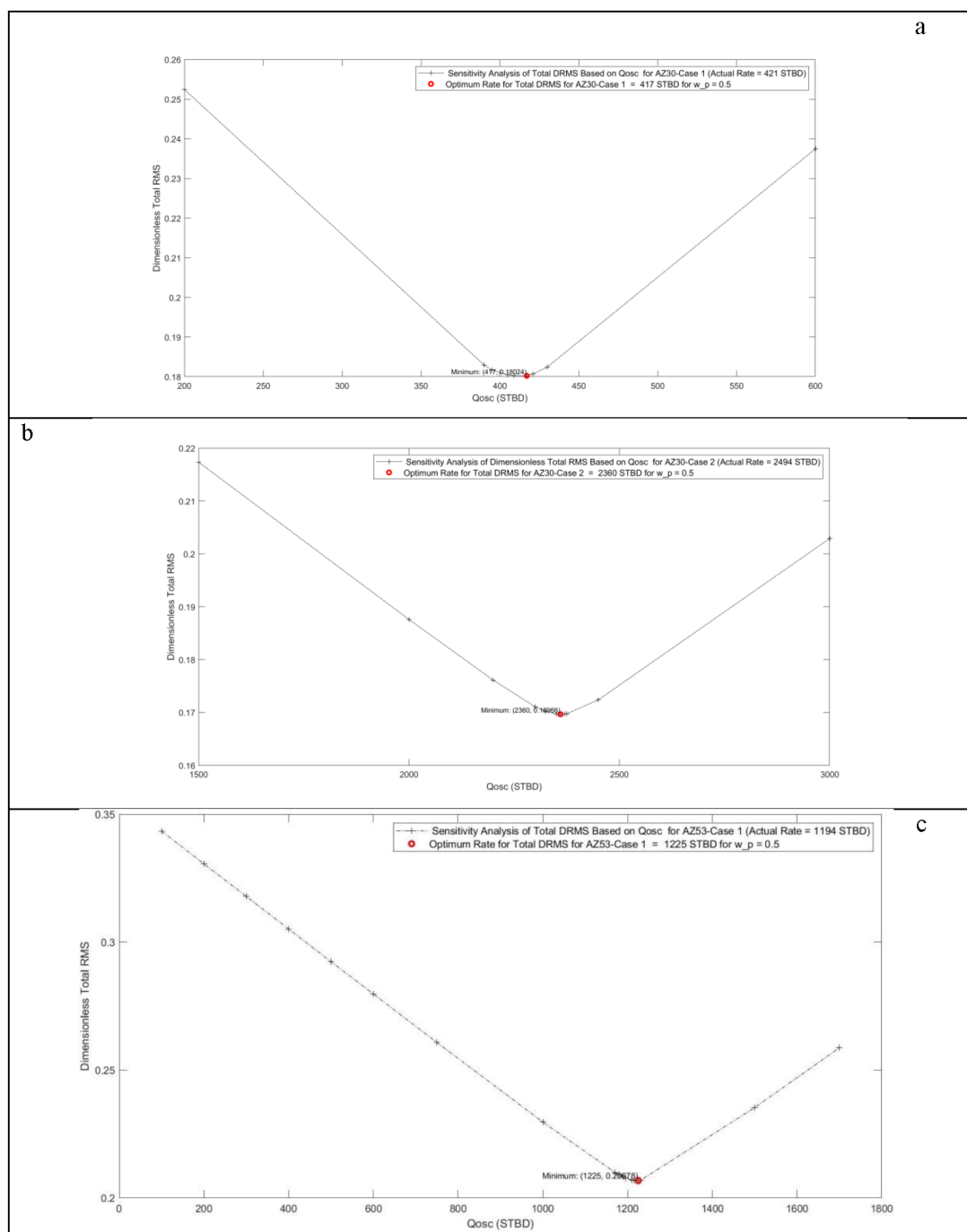


Figure 8—(a,b,c) Total dimensionless RMSE for various flow rates in for Well AZ30-Case1, AZ30-Case2, AZ53-Case1 based on the combination of dimensionless temperature and pressure assuming constant temperature in the well column.

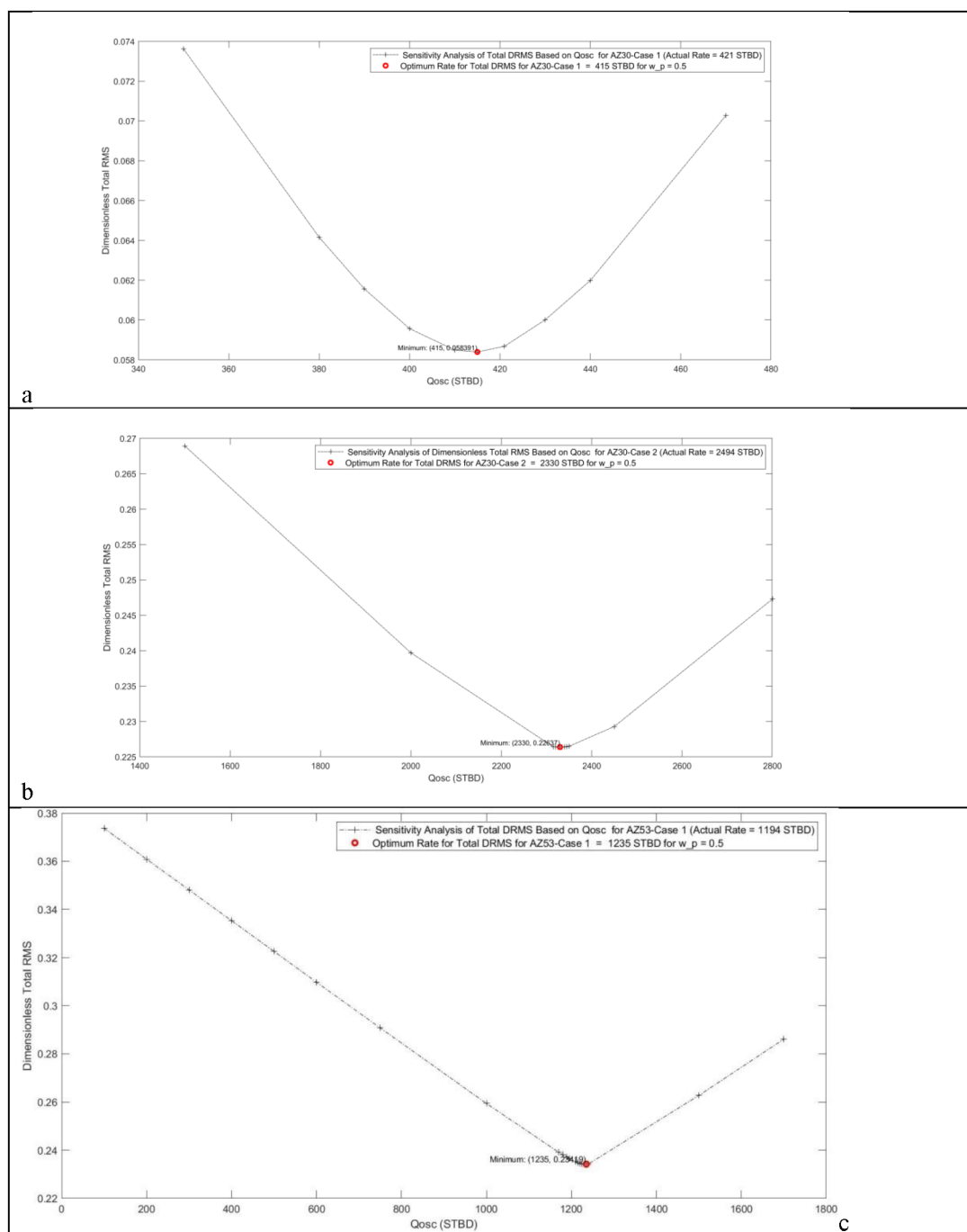


Figure 9—(a,b,c) Total dimensionless RMSE for various flow rates in for Well AZ30-Case1, AZ30-Case2, AZ53-Case1 based on the combination of dimensionless temperature and pressure assuming geothermal gradient.

Tables 2–4 compare errors across different modeling approaches. While combining pressure and temperature yields strong performance, it offers limited improvement over using pressure alone due to scale mismatches. Introducing a dimensionless approach improves the balance between variables, allowing better control over their influence in the model, even if it doesn't directly reduce flow rate error.

The numerical approach using pressure and temperature data was assessed under both constant temperature and geothermal gradient assumptions. While acceptable, it underperforms compared to detailed temperature modeling, especially in shut-in conditions where measured temperatures may not reflect actual flow behavior. In the analyzed cases, flow rate differences were minimal due to the heavy oil and black oil model used, but for lighter fluids, accurate temperature modeling becomes more critical. Among the six

methods evaluated, the combined pressure and temperature approach performed best. However, pressure data alone generally yield more accurate and reliable flow rate estimates due to their direct relationship with flow rate changes, stronger correlation with flow dynamics, and superior sensor reliability. This makes pressure-based calculations less affected by external factors and more robust than temperature-based methods, which depend heavily on uncertain thermal properties and heat transfer mechanisms.

At shallow well depths, temperature measurements are often affected by surface conditions like daily and seasonal variations, wind, and weather, causing errors and fluctuations. As a result, pressure-based flow rate calculations tend to be more accurate. When significant differences occur between pressure- and temperature-based results, the methods, sensor accuracy, and well data should be carefully reviewed. Using pressure data as the primary source, with temperature data for validation or calibration, provides the most reliable estimates. Each method has strengths and limitations, so environmental factors, tool sensitivity, and fluid properties must be considered when choosing an approach. Overall, pressure-based methods show greater reliability and accuracy due to the direct link between pressure changes and flow rate, alongside superior sensor precision.

Conclusion

The numerical integration of pressure and temperature data presented here explicitly overcomes limitations inherent in previous temperature-only graphical methods, substantially enhancing flow rates prediction reliability and operational flexibility. This study evaluated various numerical modeling approaches for estimating flow rates in three case studies using pressure and temperature data. The key conclusions are as follows:

1. The combined pressure and temperature approach showed strong performance but offered accuracy comparable to the pressure-only method due to disparities in the magnitude and units of the data.
2. Employing dimensionless normalization significantly improved the comparative analysis and adaptability of the integrated numerical model, effectively balancing contributions from both pressure and temperature data.
3. Pressure-based flow rate estimations generally exhibited higher accuracy and reliability due to direct relationships with fluid dynamics and sensor precision, whereas temperature-based methods were more sensitive to measurement inaccuracies and environmental fluctuations.
4. Numerical approaches using simplified thermal assumptions, such as constant temperature or geothermal gradients, provided acceptable results but were consistently less accurate than detailed integrated numerical modeling capturing dynamic wellbore thermal conditions.
5. The choice of the best method depends on specific well conditions, required accuracy, and system characteristics. While pressure data remains the most reliable primary source for flow rate estimation, temperature data can serve as a valuable calibration tool to refine predictions.

Nomenclature

- B_o Oil formation factor, bbl/STB, (m³/Sm³)
- C_p heat capacity at constant pressure, Btu/lbm.°F, (J/kg·°K)
- d pipe or well diameter, ft, (m)
- d_{ti} inside tubing diameter, ft, (m)
- g acceleration owing to gravity, ft/sec², (m²/s)
- g_G geothermal gradient, °F/ft, (°C/m)
- $\hat{h}$ Specific enthalpy on the unit of mass, Btu/lbm, (J/Kg)
- h_a heat-transfer coefficient for annulus fluid, Btu/hr.ft².°F, (W/m².°K)
- h_f heat-transfer coefficient for tubing fluid, Btu/hr.ft².°F, (W/m².°K)

k	conductivity, Btu/hr.ft.°F, (W/m.°K)
k_{an}	thermal conductivity of the annular fluid, Btu/hr.ft.°F, (W/m.°K)
k_c	casing-material conductivity, Btu/hr.ft.°F, (W/m.°K)
k_{cem}	conductivity of cement, Btu/hr.ft.°F, (W/m.°K)
k_e	earth or formation conductivity, Btu/hr.ft.°F, (W/m.°K)
k_t	tubing conductivity, Btu/hr.ft.°F, (W/m.°K)
$\dot{m}$	mass rate of tubing fluid, lbm/hr, (Kg/hr)
p	pressure, psia, (pa)
$\dot{Q}$	heat transfer rate of wellbore, Btu/hr, (W)
Q_v	Volumetric oil flow rate at standard condition, STB/hr, (Sm ³ /hr)
R_a	annulus resistance, hr.ft.°F/Btu, (m·°K /W)
R_c	casing resistance, hr.ft.°f/btu, (m·°K /W)
R_{cem}	cement resistance, hr.ft.°f/btu, (m·°K /W)
R_e	earth resistance, hr.ft.°f/btu, (m·°K /W)
R_f	tubing fluid resistance, hr.ft. °F/Btu, (m·°K /W)
R_{fti}	inside tubing fouling resistance, hr.ft ² .°F/Btu, (m ² ·°K /W)
R_t	total resistance, hr.ft.°F/Btu, (m·°K /W)
r_{ci}	inside radius of casing, ft, (m)
r_{co}	outside radius of casing, ft, (m)
r_{ti}	Inside tubing radius, ft, (m)
r_{to}	Outside tubing radius, ft, (m)
r_{wb}	wellbore radius, ft, (m)
T	temperature, °F, (°C)
T_{ei}	formation temperature at initial condition or at any radial distance, °F, (°C)
T_{fti}	Inside tubing fouling temperature, °F, (°C)
T_f	tubing fluid temperature, °F, (°C)
T_{wb}	wellbore temperature, °F, (°C)
t	production or injection time, hour
U_{to}	overall heat-transfer coefficient based on outside tubing fluid, Btu/hr.ft ² .°F, (W/m ² .°K)
u_t	total thermal energy, Btu/lbm, (W/Kg)
V	fluid velocity, ft/hr, (m/hr)
$\dot{W}$	work, Btu/hr, (W)
z	vertical well depth, ft, (m)
α	earth thermal diffusivity, ft ² /hr, (m ² /hr)
μ	viscosity, lb/ft.hr, (Kg/m.hr)
μ_J	Joule-Thompson coefficient, °F/psi, (°C/pa)
ρ	fluid density, lbm/ft ³ , (Kg/m ³)
v	specific volume, ft ³ /lbm, (m ³ /Kg)
θ	temperature changes, °F, (°C)

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